For discussion
on 18 July 2025

Legislative Council Panel on Transport

Implementation of Smart Mobility Initiatives

PURPOSE

This paper provides Members with an update on the latest developments in various smart mobility initiatives, focusing on "Smart Transport Infrastructure", "Data Sharing and Analytics", and "Technology Application and Services".

BACKGROUND

2. In December 2017, the HKSAR Government released the first "Smart City Blueprint for Hong Kong", introducing a range of measures across six smart domains - "Smart Mobility", "Smart Living", "Smart Environment", "Smart People", "Smart Government", and "Smart Economy" - to address urban management challenges and enhance citizens' lives through innovative technology. Subsequently, the Government published the "Smart Mobility Roadmap for Hong Kong" in July 2019, followed by the "Smart City Blueprint for Hong Kong 2.0" in December 2020. In these updates, the Transport Department (TD) expanded the Smart Mobility section by introducing new measures, such as establishing the Smart Traffic Fund (the Fund) and promoting the trial and application of autonomous vehicles (AV). These initiatives aim to optimize traffic management, improve efficiency, and advance research and innovation in vehicle-related technology.

UPDATE ON SMART MOBILITY INITIATIVES

3. The Government has been actively advancing various smart mobility initiatives by leveraging technologies such as innovative sensing and detection, artificial intelligence, and big data analytics. These efforts aim to address the growing demands of transportation and traffic, optimize road use efficiency, and provide citizens with a more convenient and seamless travel experience. Through the integration of "Smart Transport

Infrastructure", "Data Sharing and Analytics", and "Technology Application and Services", significant progress has been made across multiple projects. Key focus areas include AV technologies, advanced vehicle technologies, automated parking systems, and electronic licensing initiatives. The latest developments and work plans for these smart mobility measures are outlined in the following paragraphs.

(I) Smart Transport Infrastructure

4. Smart transport infrastructure is a fundamental pillar of smart mobility. It enhances road facilities through innovative technologies and strengthens technical frameworks to collect, analyse, and process real-time traffic data. This enables the dissemination of traffic information to road users, optimising travel planning. The Government has been actively advancing Hong Kong's smart transport infrastructure through various initiatives, including promoting AV trials and applications, implementing real-time adaptive traffic signal systems, advancing smart motorways, and upgrading traffic control and surveillance systems.

Trial and Application of Autonomous Vehicles

5. One of the key smart mobility measures outlined in the "Smart Mobility Roadmap" and "Smart City Blueprint 2.0" is to keep pace with the technological and industry advancements in vehicle-to-everything (V2X) and AV. The Government completed the legislative amendment process for the "Road Traffic (Amendment) (Autonomous Vehicles) Ordinance 2023" and the "Road Traffic (Autonomous Vehicles) Regulation" (Cap. 374AA) (AV Regulation) in May 2023 and January 2024, respectively. The new AV regulatory framework was subsequently implemented in March 2024, alongside the TD's release of the "Code of Practice for Trial and Pilot Use of Autonomous Vehicles". Under this framework, TD will issue "Pilot Licence" and "AV Certificate" to "Pilot Proprietor" participating in the AV "Pilot Scheme". Based on the local traffic conditions and input from key stakeholders, pilot proprietors will be permitted to trial or operate one or more approved AVs in designated areas. This approach provides the industry with greater flexibility to test and deploy AV under more complex road conditions while maintaining road safety.

6. Several AV projects are currently in progress. As of the end of June 2025, the TD has issued "Movement Permits" for 21 AV to undergo trials at 12 different locations. At present, six AV are being trialed at four

locations, including Tai Po Science Park, Fairview Park, the Hong Kong Productivity Council, and Tuen Mun Lung Mun Road EcoPark. These projects have demonstrated that trained autonomous driving systems can adapt to Hong Kong's traffic patterns, with relevant technologies proving extendable for AV trials on public roads. TD will actively engage with project stakeholders regarding post-trial implementation. Additionally, under the new AV Regulation enacted in 2024, TD has issued two AV Pilot Licences - approving 30 AV for trial in North Lantau and two autonomous light buses for trial in the West Kowloon Cultural District (WKCD).

7. The North Lantau project, launched in December 2024, has made significant progress. Under the new regulatory framework, the TD has streamlined the approval process while maintaining rigorous oversight, actively facilitating licencees and granting approvals. Within months, several technical advancements have been achieved, including scaling the trial from a single vehicle to 10 vehicles operating simultaneously to collect multi-vehicle data, conducting passenger-carrying trials on specific routes following safety assessments, expanding the trial scope to more complex roads, and increasing the speed limit to 50 km/h based on the trial route's maximum speed limit. Meanwhile, the WKCD project will trial autonomous light buses equipped with an autonomous driving system designed to comply with Hong Kong's left-hand traffic rule. The research institute plans to conduct passenger-carrying trials on public roads within WKCD by the end of this year.

8. Looking ahead, the Government remains committed to prioritizing safety while leveraging project insights to refine technical standards and steadily advance AV development. In addition to the aforementioned initiatives, we are actively progressing other projects, including autonomous light buses operating between Hong Kong Science Park in Sha Tin and MTR University Station, as well as an autonomous transportation system connecting SKYCITY at Hong Kong International Airport with the Hong Kong Port Island of the Hong Kong-Zhuhai-Macao Bridge. The goal is to implement autonomous driving services in local communities within the next one to two years.

Real-time Adaptive Traffic Signal System

9. As real-time adaptive traffic signal technology continues to mature, its applications have expanded globally. Many cities in the Mainland are actively integrating this technology, using artificial intelligence to analyze traffic conditions and adjust signals in real time to optimize timing for signalised junctions, effectively reducing road

congestion. The TD successfully completed the pilot scheme for the "Real-time Adaptive Traffic Signal System" at five independent signalised junctions as early as 2021. Additionally, in 2022, TD began installing and trialing the system at eight linked signalised junctions in Tung Chung town center. The trial was completed in 2024 and is currently operating smoothly. Data indicates that following the installation of the system at the tested junctions, the average waiting time for vehicles passing through was reduced by approximately 5-10%, significantly improving the operational efficiency of the junctions.

10. Building on previous successes, the TD is expanding the system to approximately 50 suitable independent signalised junctions across Hong Kong. The project secured funding approval from the Finance Committee in June 2024, and the related consultancy study commenced in July 2024. System installation is expected to begin in 2026, followed by testing and gradual deployment. The entire project is anticipated to be completed by 2027.

11. Meanwhile, the TD is conducting further trials of the system at busier linked signalised junctions (such as Wan Chai North and Causeway Bay). At the same time, it is exploring the use of other relevant technologies (including those from the Mainland), to support the broader application of the system at suitable junctions. As part of the Government's development of new urban areas, TD will collaborate with the relevant departments overseeing infrastructure projects to integrate the system into signalised junctions from the outset. For instance, the first batch of systems in the Tung Chung New Town Extension is expected to be installed by 2026.

Traffic Control and Surveillance Systems

12. To ensure the safe and efficient operation of Hong Kong's strategic roads, all government tunnels and major trunk roads are equipped with Traffic Control and Surveillance Systems (TCSS), which operate continuously. These systems include lane control signals, variable message signs, closed-circuit televisions (CCTVs), and automatic incident detection systems to support real-time traffic management and monitoring. Government departments rely on TCSS for 24-hour surveillance of road traffic in tunnels and major trunk roads, enabling swift response to incidents by notifying tunnel staff and the Police. Meanwhile, lane control signals and variable message signs help direct motorists to reduce speed or take alternative routes, ensuring road user safety and minimizing the broader impact of traffic disruptions.

13. Several tunnels and major roads' TCSS have been in operation for many years and have exceeded their serviceable lifespan. To mitigate the risk of malfunction, the Government consulted the Panel on Transport in May this year and is now preparing to seek funding approval from the Legislative Council for the replacement and renewal of these systems. The upgraded and renewed TCSS will incorporate advanced technologies and new features, including active warning systems (installed at the Western Harbour Crossing and Cross-Harbour Tunnel); CCTVs and automatic incident detection systems with higher resolution digital cameras; and Building Information Modelling data repository[1] to enhance smart tunnel management.

14. These new features will significantly enhance the efficiency of traffic monitoring and management. They will also support dedicated pilot projects (see paragraph 15 below) aimed at accelerating the innovative development of smart motorway management.

Smart Motorway Pilot Scheme at Ting Kau Bridge

15. "Moving Towards the Application of Smart Motorway Management" is one of the key initial recommendations of the "Traffic and Transport Strategy Study". This initiative aims to leverage technology to maximize the use of limited road resources and strengthen the resilience of transport infrastructure in responding to incidents. To support this effort, the TD launched a Pilot Scheme on 29 December 2024, at Ting Kau Bridge southbound to test smart motorway technology and refine detailed operational arrangements.

16. From the commencement of the Pilot Scheme until the end of June this year, a total of 19 minor traffic incidents occurred in the designated road section. Of these, 13 were successfully detected within 10 seconds[2] by the automatic incident detection system, which simultaneously issued alerts to the operational staff of the Tsing Ma Control Area. This enhancement has significantly improved the efficiency of traffic incident management and eased traffic flow.

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[1] The Building Information Modelling (BIM) data repository allows maintenance staff to efficiently and comprehensively understand the configurations of various new equipment and systems using computer support. It helps trace the causes of facility damage, identify other potentially affected equipment, and accelerate repairs while optimizing maintenance programs.

[2] For the remaining six incidents, detection times were prolonged due to external environmental factors affecting individual equipment. A video analysis of these cases has been conducted to fine-tune system parameters accordingly. Moving forward, we will continue monitoring and optimising system performance to enhance detection accuracy and response efficiency.

Compared to the previous operation mode, which primarily relied on periodic inspections by operational staff using CCTV, the implementation of the Pilot Scheme has reduced the time required for staff to identify traffic incidents by nearly 90%. The TD will continue monitoring the scheme and conduct an interim review based on data collected in the first half of 2025 to assess its effectiveness. The insights gathered from the Pilot Scheme will also inform strategies for the future implementation of smart motorways, including further trials on other suitable major trunk roads. Additionally, the Government will integrate smart motorway elements and requirements into related projects, such as road widening initiatives or the replacement of TCSS on existing major trunk roads.

(II) Data Sharing and Analytics

17. As smart transport infrastructure projects advance, the resulting traffic data has become essential for enhancing transport system planning, management efficiency, and operational resilience. The Government actively leverages data analytics to gain insights into citizens' travel habits and transport mode trends, supporting more effective transport planning and traffic management. Additionally, traffic data is made accessible through various platforms, including the TD's "HKeMobility" mobile application and the "Traffic Data Analytics System", which provide real-time and forecast traffic data alongside public transport information. These initiatives foster the development of more innovative and personalized services, facilitate citizens' journey planning and public transport use, and promote cross-sector applications in smart mobility.

<u>"HKeMobility" Mobile Application</u>

18. Since its 2021 update, the TD has enhanced the "HKeMobility" mobile application to provide personalized traffic and public transport information, including expanded real-time arrival data. These features help citizens estimate journey times, make informed travel decisions, and choose optimal departure times. TD has continuously improved the app's functionality by integrating real-time toll information for Tai Lam Tunnel and road harbor crossings, queue ticket distribution and calling status for TD's licensing offices, and an enhanced user interface. Most of the data available on "HKeMobility" is also shared via the Government's "Open Data Portal" (data.gov.hk) and the "Common Spatial Data Infrastructure Portal" (portal.csdi.gov.hk). Since its launch, the mobile application has gained widespread adoption, with over 2.84 million downloads and an average daily hit rate of 190 000 as of June 2025. Moving forward, we

will actively explore the feasibility of integrating electronic services and collaborate with the Digital Policy Office to advance the Government's goal of establishing "single-portal access" for public services.

Installation of Information Display Panels at Sheltered Bus Stops and Covered Public Transport Interchanges

19. To enhance passenger convenience and help them better manage waiting times and travel plans, the Government has been subsidising franchised bus companies on a matching basis since 2016 to accelerate the installation of real-time bus arrival information display panels at sheltered bus stops where technically feasible (including locations with power supply and sufficient space). As of June 2025, franchised bus companies have installed information display panels at 1 114 sheltered bus stops and Government-owned covered public transport interchanges. The TD and franchised bus companies will continue to assess the feasibility of expanding the installation of real-time bus arrival information display panels at additional sheltered bus stops as appropriate.

(III) Technology Application and Services

20. By integrating smart transport infrastructure with relevant data, the Government leverages technology to enhance traffic management measures and policies. These initiatives aim to improve transport efficiency, promote road safety, and provide greater convenience to citizens. Key projects include the adoption of automated parking systems, the expansion of advanced vehicle technology applications, the implementation of traffic e-enforcement system, and the introduction of electronic licensing initiatives to digitize transport services and enhance the overall travel experience.

Automated Parking System

21. The adoption of automated parking systems enhances space utilization and increases parking density, helping to address public demand for parking spaces. The Government is actively promoting these systems in suitable public and short-term tenancy car parks while encouraging their adoption in private car parks. Regarding short-term tenancy car parks, the Government has launched three automated parking system projects[3], providing approximately 180 automated parking spaces. The automated

[3] Located at Hoi Shing Road in Tsuen Wan, Pak Shek Kok in Tai Po, and Tung Chau Street in Sham Shui Po respectively.

parking system at Hoi Wang Road short-term tenancy car park in Yau Ma Tei is expected to be operational in 2025. Additionally, short-term tenancy car parks at Po Tai Street in Ma On Shan and Lok Wah Street in Tsz Wan Shan will also be equipped with automated parking systems, with the tender process scheduled for 2025.

22. The Government plans to provide at least 3 000 automated parking spaces by 2026. Currently, five public automated car park projects are under construction, incorporating various types of automated parking systems (including puzzle stacking, vertical lifting and horizontal sliding)[4]. These projects are expected to be completed between the end of this year and 2029. Additionally, the Hong Kong Airport Authority is gradually completing automated car parks on the Hong Kong Port Island of the Hong Kong-Zhuhai-Macao Bridge to accommodate transit and visiting travelers from Guangdong and Macau. To promote wider adoption of automated parking systems in Hong Kong, private developers' participation is crucial. The Electrical and Mechanical Services Department and the TD have issued industry guidelines on automated parking systems. Moving forward, TD will actively encourage developers to implement these systems and explore additional incentive measures.

Advanced Vehicle Technologies

23. The TD regularly reviews and updates regulations and guidelines, referencing international standards to align with the latest automotive market developments. Earlier this year, the Government introduced legislative amendments to optimize vehicle design and construction standards, facilitating the adoption of new automotive technologies in Hong Kong to enhance road safety and convenience. Since March 2025, drivers have been permitted to use remote control parking functions and access non-driving-related information on in-vehicle visual display units.

24. Existing regulations and approvals from the TD allow vehicle manufacturers to equip vehicles with Level 2 advanced driver assistance systems (ADAS), which are widely used in the market. These systems include remote control parking, non-driving-related information in visual display units, automatic parking, vehicle side collision warning, automatic anti-skid, assisted direction guidance, automatic lane changing, lane

[4] They are Joint-user Government Office Building in Area 67 of Tseung Kwan O, the District Open Space, Sports Centre cum Public Vehicle Park at Sze Mei Street in San Po Kong, the Open Space with Public Vehicle Park at Yen Chow Street West in Sham Shui Po, and the Amenity Complex in Area 103 of Ma On Shan and Town Park with Public Vehicle Park in Area 66 of Tseung Kwan O.

keeping, cruise control, and automatic emergency braking, etc.. Manufacturers must ensure these features comply with recognized national or international standards and are used within specified driving environments. TD will continue working with the industry to facilitate the introduction of more advanced vehicle technologies.

Traffic e-Enforcement System

25. To enhance the efficiency of traffic enforcement through technology and provide more convenient public services, the Government completed legislative amendments for electronic traffic enforcement in 2024. These amendments enable the Hong Kong Police Force to issue fixed penalty notices (e-tickets) for traffic violations electronically. Since November 18, 2024, the TD has been collecting verified electronic contact details from registered vehicle owners, driving licence holders, and permit holders to facilitate the issuance of e-tickets. Beginning 15 June 2025, the Police will progressively implement the e-ticket system and simultaneously launch the "Electronic Traffic Ticket Platform" website (www.etrafficticket.gov.hk) and mobile application. These platforms will allow citizens to access their e-tickets and conveniently pay fines using newly introduced electronic payment methods.

Electronic Licensing Initiatives

26. To support the Government's "Be the Smart Regulator" initiative and address the growing demand for licensing services, the TD has introduced electronic licensing measures and simplified application processes. Currently, TD offers over 20 online services (including high-demand services such as vehicle licence renewals, full driving licence renewals, and international driving permit (IDP) applications) allowing citizens to submit applications via mobile devices or computers using "iAM Smart" digital signatures and/or e-Form functions, eliminating the need to visit licensing offices and wait for counter services. TD plans to further expand these services by adding an additional 20 licensing options, including applications for excess dimension permits and Certificates of Driving Licence Particulars.

27. The TD has been progressively rolling out various electronic licensing initiatives to improve operational efficiency, enhance public convenience, and reduce congestion and wait times at licensing offices. These initiatives include the enhanced Vehicle Licence (eVL), the "E-

Licensing Portal" one-stop platform[5], IDP Self-Service Kiosks[6], "e-Auction"[7] and the electronic driving licence (eDL). Launched in December 2024, the eVL benefits over 800,000 vehicle owners by eliminating the need to replace the paper vehicle licence upon renewal after initially obtaining a new licence without a printed expiry date. Additionally, TD has simplified the required application documents and implemented artificial intelligence technology to assist in processing online vehicle licence renewals, aiming to reduce processing times of applications that do not require manual verification of supporting documents from 10 working days to just 3. As application and processing procedures become increasingly automated, more citizens are expected to apply for vehicle licences online, avoiding the need for in-person visits to TD licensing offices. Furthermore, following the passage of relevant amendment regulations by the Legislative Council on 12 June, TD will introduce eDL in September this year, allowing licence holders to choose to display their eDL on smartphones. This measure is expected to provide convenience to approximately 2.5 million licence holders. Meanwhile, the Government plans to submit legislative amendments regarding the International Circulation Permit to the Legislative Council within this year to streamline application and processing procedures, facilitating electronic applications for the applicants.

Smart Traffic Fund

28. To support the three pillars of smart mobility, the Government established the Smart Traffic Fund, earmarking HK$1 billion to subsidise local enterprises and organizations conducting research and applying vehicle-related innovations and technologies. The initiative aims to enhance commuting convenience, optimize road network efficiency, and improve driving safety. Launched in March 2021, the Fund began

[5] Launched in December 2024, the "e-Licensing Portal" provides a one-stop platform for licence applications and inquiries. It features a user-friendly dashboard that clearly displays information about registered users' licences, expiry dates, and the status of applications. This online portal enables members of the public to conveniently manage their licences and vehicles while offering the option to receive email reminders for driving licence and vehicle licence renewals.

[6] Since January 2025, "IDP Self-Service Kiosks" have been installed at the Hong Kong Licensing Office and the Kowloon Licensing Office to provide the public with a convenient option for applying for an International Driving Permit (IDP). These kiosks support online booking and allow applicants to collect their permits within half an hour after completing the application process, provided they have all the necessary documents.

[7] Launched in December 2024, the "e-Auction" platform enables bidders to participate in auctions for Ordinary Vehicle Registration Marks remotely using computers or mobile devices. This one-stop platform supports electronic payments and streamlines follow-up procedures, providing a convenient and efficient bidding experience.

accepting applications and fostered collaboration among the Government, industry, academia, and research sectors to drive long-term smart mobility development in Hong Kong. The Fund's Management Committee has been overseeing applications and monitoring its overall operations. As of June 2025, the Management Committee has approved 76 project applications, with total funding of approximately HK$541 million. This includes HK$170 million allocated to five AV testing projects and nine V2X and related technology initiatives, HK$138 million for 21 projects focused on enhancing road safety and improving driving behavior, and HK$102 million for 16 projects aimed at improving public transport safety and scheduling.

WAY FORWARD

29. Looking ahead, the Government will continue to actively promote smart mobility by consolidating existing strategies, drawing on global experiences, adopting appropriate new technologies, and launching projects that enhance convenience for road users. The TD initiated the Traffic and Transport Strategy Study at the end of 2021 to develop a long-term Transport Strategy Blueprint for Hong Kong. Nine initial recommendations were introduced in December 2023, and TD has been progressively implementing smart mobility pilot schemes since 2024. In collaboration with relevant policy bureaux and departments, we aim to deepen various studies and finalize the Transport Strategy Blueprint this year to maintain a reliable, safe, smart, environmentally friendly, and highly efficient transport system for Hong Kong.

ADVICE SOUGHT

30. Members are invited to note the above updates on smart mobility initiatives.

Transport and Logistics Bureau
Transport Department
July 2025